

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460659.7771 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.095 +/- 0.04
Transit depth (Rp/Rs)^2	0.9 +/- 0.75 [%]
Orbital Inclination (inc)	86.72 +/- 1.7
Airmass coefficient 1 (a1)	0.966 +/- 0.015
Airmass coefficient 2 (a2)	0.02 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.61 %
Best Comparison Star	#1 - [452, 410]
Optimal Aperture	3.41
Optimal Annulus	16.36
Transit Duration (day)	0.1089 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.095 ± 0.04 vs 0.1489 (deviation 1.35σ)
Duration fit vs published	0.1089 d vs 0.1304 d (deviation 3.91σ)
Mid-time O–C	9.3 min
Depth SNR	2.4
Residual scatter	1.61 %

Light curve

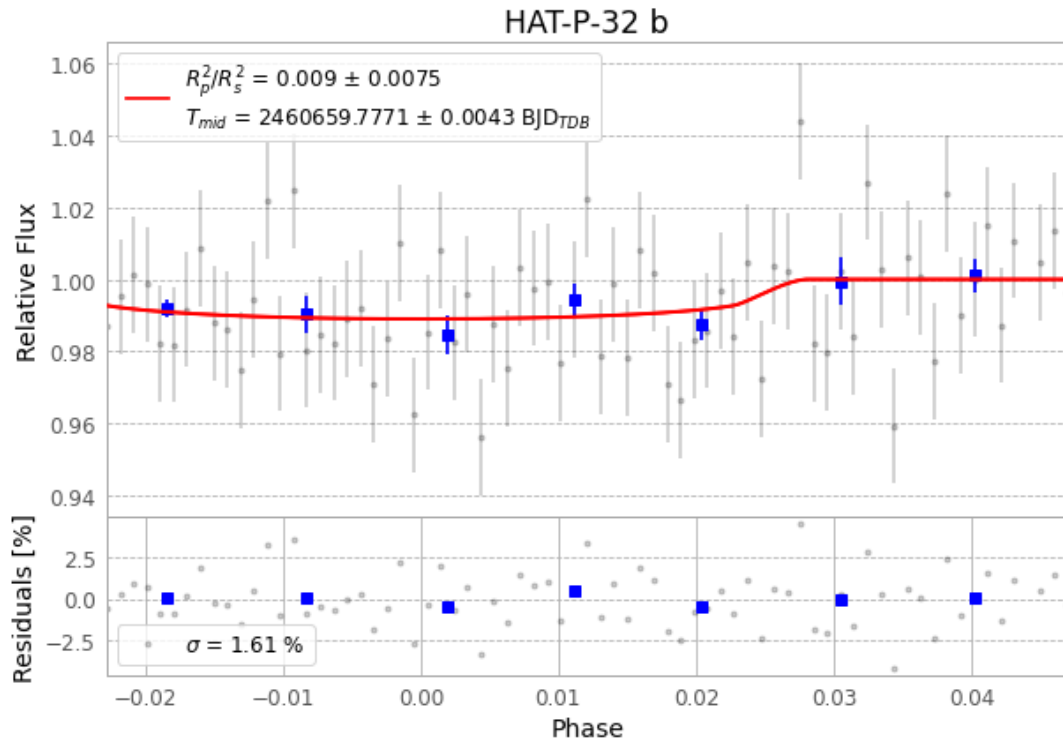


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-12-14.png (SHA-256 06032d083591f791...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 275], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 53924411b55bbab0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1965f5e

Data provenance

73 FITS frames (48 MB), HATP-32241215052115.FITS → HATP-32241215085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241215.log (893403fe07be...), FinalLightCurve_HAT-P-32 b_2024-12-14.png (06032d083591...), FinalLightCurve_HAT-P-32 b_2024-12-14.csv (d8e4e46bdc46...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 22:54Z.